SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA1021	**Course Title:**CSA1021 SOFTWARE ENGINEERING	Software Engineering
CO Assessed:	CO1	**Assessment:C01_Case Study Analysis**	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
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Date:15/10/2026		**Duration:16/10/26**	60 minutes

Code of Conduct

I _____Rashmitha Senthil Kumar (192525034)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ___Rashmitha .S_____

Annexure A – Case Study Analysis

Instructions to Students:

Each student will be allotted **one problem statement**. Analyze the assigned problem systematically by following the steps given below. Your report should demonstrate your understanding of software engineering concepts and your ability to design an appropriate software solution.

Based on your allotted problem statement, prepare a report by answering the following questions:

1. **Understand the Problem Statement**
   - Explain the problem in your own words.
   - Identify the objectives of the proposed system.
   - State the scope of the problem.
2. **Identify Key Stakeholders and Challenges**
   - Identify all the stakeholders involved in the system.
   - Explain the roles and responsibilities of each stakeholder.
   - Discuss the major challenges and issues faced by the stakeholders.
3. **Analyze the Current Approach**
   - Describe the existing system or current process.
   - Identify its strengths and limitations.
   - Analyze the problems or gaps that need to be addressed.

4. **Propose a Suitable Solution Using Software Engineering Principles**
    - Design a software-based solution for the given problem.
    - Describe the major functional modules of the proposed system.
    - Identify the functional and non-functional requirements.
    - Explain how software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security are incorporated into your solution.
5. **Justify the Chosen Methodology/Framework**
    - Select an appropriate Software Development Life Cycle (SDLC) model or software development framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).
    - Justify why the selected methodology/framework is suitable for the proposed system.
    - Explain how it supports successful project development and maintenance.
6. **Discuss Expected Outcomes and Limitations**
    - Describe the expected benefits of the proposed solution.
    - Explain how the solution addresses the identified challenges.
    - Discuss potential risks, implementation challenges, and limitations.
    - Suggest possible future enhancements.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding (30%)	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.
Stakeholder & Challenge Analysis (25%)	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.
Current System Analysis (20%)	Thoroughly analyzes the existing system, strengths, weaknesses, and identifies all gaps.	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.	Proposed solution is unclear or inappropriate; requirements are largely missing.
Proposed Software Solution & Software Engineering Principles (15%)	Proposes an innovative, feasible solution applying appropriate software engineering principles (modularity, scalability, maintainability, security, etc.).	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.	Methodology is inappropriate, poorly justified, or not discussed.
Methodology/ Framework Justification (10%)	Selects the most suitable SDLC/model/framework and provides strong technical justification.	Suitable methodology selected with adequate justification.	Methodology selected but justification is weak.	Outcomes, limitations, or future enhancements are missing; report lacks organization and clarity.

Criteria	**Mark**
Problem Understanding (30%)	/5
Stakeholder & Challenge Analysis (25%)	/5
Current System Analysis (20%)	/5
Proposed Software Solution & Software Engineering Principles (15%)	/5
Methodology/ Framework Justification (10%)	/5
TOTAL	/25

Industry Context: Urban water distribution networks lose billions of gallons of clean drinking water annually due to hidden pipe bursts and slow sub-surface leaks. If an underground pipe burst goes unnoticed for hours, it can wash out roads flood basements, and waste expensive treated water. Municipal water teams need an automated network that tracks aqueduct pressure variations to pinpoint leak locations and isolate broken lines instantly.

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Problem Statement: Design and develop a fault-tolerant Municipal Aqueduct Leak Detection Platform that monitors water pressure changes and coordinates valve systems across city grids. The platform must ingest high-velocity streaming data—including acoustic pipe sensors, water flow meters, valve statuses, and

soil moisture indicators—from thousands of utility nodes simultaneously. It must isolate breaks automatically.

Objectives:

1.Ingest and process telemetry from low-power aqueduct sensor networks with zero data loss.

2.Pinpoint underground leak positions using acoustic wave travel time difference checks.

3.Execute automated isolation commands to remote water valves when a pressure drop is verified.

4.Balance city water pressure targets dynamically to prevent pipes from cracking during demand drops.

5.Maintain unalterable, audit-ready records of city water utilization profiles for utility boards.

Expected Outcomes:

1.Significant reduction in clean water losses across urban distribution infrastructure.

2.Near-zero street flooding risks from large, uncontained underground water pipe bursts.

3.Lower maintenance costs via fast, automated pinpointing of fragile water lines.

4.Safe, steady water pressure delivery for city homes and commercial sectors.

FAULT-TOLERANT MUNICIPAL AQUEDUCT LEAK DETECTION PLATFORM

1. Understand the Problem Statement:

1.1 Problem Explanation:

An urban water distribution network carries treated drinking water through interconnected underground pipes. Its operation depends on maintaining suitable pressure, detecting damaged sections, and controlling the valves that direct water through different areas. When a pipe develops a hidden leak, water may escape continuously without immediately reaching the surface. A major burst can also damage nearby roads and buildings before maintenance teams arrive.

The challenge is more complex than identifying low pressure. Pressure can decrease because of legitimate consumption, pump switching, maintenance, or a pipe failure. Acoustic sensors may detect background vibrations, while soil moisture can increase because of rainfall. Decisions based on one measurement can produce false alarms and unnecessary service interruptions.

The proposed platform combines pressure, acoustic, flow, moisture, and valve-status information. It continuously evaluates this evidence, estimates the damaged pipe section, and coordinates approved isolation actions. It must remain dependable when communication links, devices, or computing services fail. The core problem is to convert large volumes of uncertain sensor data into timely, explainable, and verified operational decisions.

1.2 Objectives:

• Dependable ingestion: Receive concurrent telemetry from thousands of nodes and recover buffered measurements after temporary communication failures.

• Reliable detection: Identify abnormal pressure and flow while distinguishing likely leaks from normal operational changes.

• Leak localization: Estimate underground leak positions using acoustic arrival-time differences and verified pipe information.

• Controlled isolation: Automatically isolate confirmed breaks when approved safety and confidence conditions are satisfied.

• Pressure management: Coordinate bounded pressure adjustments as demand changes.

• Operational continuity: Continue essential monitoring during defined component failures.

• Accountability: Preserve traceable, tamper-evident utilization, incident, configuration, and valve-action records.

1.3 Success Criteria:

Evaluate detection delay, verified isolation time, false-alarm frequency, location error, telemetry recovery, continuity during service failures, and completeness of incident reconstruction. Acceptance thresholds require utility approval. These are proposed evaluation criteria, not achieved results.

1.4 Scope:

Included activities: Device registration, telemetry collection, validation, leak detection, acoustic localization, valve coordination, bounded pressure management, incident tracking, maintenance updates, and utilization reporting. The platform maintains relationships between pipes, sensors, valves, and service zones. Initial implementation covers a pilot zone before wider expansion.

Excluded activities: Physical excavation, pipe replacement, drinking-water quality certification, and unrestricted redesign of pumping infrastructure. The platform supports field teams but does not replace physical inspection or hydraulic engineering.

1.5 Assumptions and Constraints:

Participating devices need unique identities, and controllable valves need usable status feedback; these capabilities must be verified. Localization requires suitable sensor placement, synchronized observations, and a defensible propagation model. Incorrect maps, deteriorated valves, restricted connectivity, and battery capacity constrain performance.

“Zero data loss” is an objective within a defined recovery envelope. Data cannot be recovered if a device fails before recording it or an outage exceeds storage capacity. Automatic isolation reduces flooding exposure but cannot eliminate every consequence of a burst.

2. Identify Key Stakeholders and Challenges:

2.1 Stakeholders and Responsibilities:

Stakeholder	Responsibility and primary concern
Utility board	Approve funding, service policies, priorities, and acceptance criteria; conserve water and maintain accountable expenditure.
Control-room operators	Supervise incidents, review uncertainty, and authorize exceptional actions; require clear alerts and reliable status.
Field maintenance teams	Inspect faults, repair assets, calibrate devices, and confirm outcomes; need accurate locations and safe conditions.
Hydraulic engineers	Define pressure limits and validate isolation plans; prevent harmful transients and unintended supply loss.
Residents and businesses	Use services and receive interruption information; depend on reliable supply and timely restoration.
Essential-service facilities	Coordinate continuity arrangements for hospitals and other priority users; protect critical supply.
IT and security teams	Maintain software, connectivity, access, backups, and recovery; protect availability and integrity.
Equipment suppliers	Support interfaces, calibration, firmware, and maintenance; enable compatible equipment integration.

2.2 Responsibility Boundaries:

Engineers approve operating rules; developers implement them; operators supervise application; field teams confirm conditions. Administrators do not automatically receive valve privileges. Separating responsibilities limits the ability of one user to alter evidence, change rules, and execute actions without oversight.

2.3 Major Challenges:

Data volume and timing: Devices transmit at different rates, and incidents can produce simultaneous traffic bursts. Processing must prioritize urgent events while retaining routine measurements.

Communication and battery limitations: Low-power sensors may not support continuous transmission of raw acoustic data. Local processing and selective transmission must balance detection, power, and bandwidth.

Uncertain evidence: Sensor drift, consumption changes, background noise, and maintenance can resemble leakage. Detection must combine evidence and expose uncertainty.

Localization accuracy: Timing analysis depends on synchronization, geometry, propagation speed, and signal quality. Branches and reflections can create several plausible locations.

Isolation safety: Incorrect valves can interrupt unaffected areas. Actuation must follow approved profiles to limit damaging pressure changes.

Fault tolerance: Failed servers, disconnected gateways, and stuck valves require defined responses. Missing information must appear as degraded visibility.

Audit integrity: Historical evidence must remain available and alterations detectable, including changes attempted by privileged users.

3. Analyze the Current Approach:

3.1 Baseline Process:

No particular municipality or installed system is identified in the question. This analysis therefore uses a defined comparison baseline rather than an invented real-world survey. It combines periodic inspections, customer complaints, meter readings, separate pressure displays, and manually coordinated field response.

The baseline sequence is: receive an abnormal reading or complaint; review records; dispatch an investigation team; identify isolation valves; close valves and repair the section; and document the incident. Existing supervisory control or remote valves should be integrated after interface and safety verification.

3.2 Strengths:

Experienced operators understand local demand patterns, and field inspection confirms physical damage. Existing devices and procedures reduce replacement needs. Human review remains valuable when records are incomplete or several explanations fit the evidence. The proposed platform retains this strength through supervised exception handling.

3.3 Limitations:

Intermittent inspection can delay recognition of hidden failures. Separate records require manual correlation, uncertain locations increase investigation effort, and manual coordination can delay isolation. Fragmented records also obstruct comparison of response times and confirmation that an action succeeded.

3.4 Gap Analysis:

Identified gap	Consequence	Proposed response
No continuous combined evidence	Hidden leaks persist before investigation.	Correlate pressure, flow, acoustic, and supporting moisture observations.
Uncertain fault location	Wider inspection and excavation effort.	Report candidate pipe sections with location confidence.
Delayed isolation coordination	Discharge continues after detection.	Execute approved automatic isolation when conditions permit.
Unverified valve commands	Isolation may be incorrectly assumed.	Compare requested and observed valve states.
Communication interruptions	Incomplete operational history.	Buffer, acknowledge, and replay retained events.

3.5 Root-Cause Analysis:

The main gap is the weak connection between observation, interpretation, action, and verification. More sensors alone will not resolve it. A pressure alarm does not explain its cause, and a valve command does not prove physical closure. The platform preserves the evidence behind a decision and checks its operational result.

4. Propose a Suitable Solution Using Software Engineering Principles:

4.1 Proposed Architecture:

Layer	Main functions
Field devices	Measure pressure, flow, acoustic activity, moisture, and valve position.
Edge gateways	Authenticate devices, buffer records, process suitable data locally, and forward telemetry.
Streaming services	Durably ingest events, validate schemas, deduplicate, and correlate observations.
Decision and control	Detect incidents, estimate locations, evaluate isolation rules, and verify actions.
Operational applications	Display conditions, coordinate maintenance, report utilization, and preserve audit evidence.

A durable event stream separates collection from processing, so a temporary analytics failure need not stop accepted measurements from being retained. Monitoring, decisions, and control have distinct interfaces. The platform respects established local equipment protections and does not bypass their responsibilities.

4.2 Core Data Records:

The principal records are Asset, Device, TelemetryEvent, Incident, ValveCommand, MaintenanceAction, and AuditEntry. An incident links supporting events, topology version, decision-rule version, selected valves, command results, and field findings. These links support operational explanation and investigation.

4.3 Functional Modules:

Device and topology management: Maintain identity, location, calibration, pipe connections, and valve relationships. Authorize and version changes.

Telemetry ingestion: Receive timestamps, sequence numbers, measurements, units, and quality flags. Identify invalid values and prevent duplicate totals.

Leak detection: Correlate pressure and flow abnormalities with acoustic evidence and known operations. Treat moisture as supporting environmental evidence.

Acoustic localization: Compare synchronized signals and map plausible arrival-time differences to candidate pipe sections.

Valve coordination: Check boundaries, protected consumers, permissions, and actuator readiness before commands.

Pressure coordination: Apply approved limits, bounded adjustment rates, and stable rules; avoid conflicting controller instructions.

Incident and maintenance management: Track acknowledgement, investigation, repair, restoration, and closure.

Audit and reporting: Produce utilization summaries and preserve operational and administrative evidence.

4.4 Acoustic Localization Logic:

For a simplified straight pipe, L is the distance between sensors A and B, c is estimated acoustic propagation speed, and x is the leak distance from A. With the convention $\Delta t = tA - tB$:

$$\Delta t = x/c - (L - x)/c \qquad \text{Therefore, } x = (L + c\Delta t)/2$$

The calculation requires valid modelling assumptions. Reject inconsistent results, account for timing uncertainty, and assess alternative paths in branched networks. Report a defensible location estimate rather than an unsupported exact excavation point.

4.5 Detection and Isolation Workflow:

1. Collect, timestamp, and validate telemetry.

2. Check freshness, device health, and missing sequences.

3. Detect abnormal pressure or flow behaviour.

4. Compare planned operations and supporting evidence.

5. Estimate the affected section and confidence.

6. Evaluate isolation boundaries and service restrictions.

7. Execute automatic isolation only when required conditions are met.

8. Verify valve feedback and subsequent measurements.

9. Escalate failed, ambiguous, or incomplete isolation.

10. Record repairs and restore service through an approved procedure.

High-confidence incidents with verified plans can trigger automatic action. Ambiguous events or invalid maps require review, preserving automation without unrestricted control.

4.6 Functional Requirements:

ID	Requirement	Acceptance evidence
FR1	Register devices against network assets.	Reject unknown or unauthorized device identities.
FR2	Recover retained telemetry after outages.	Reconcile recovered and missing sequences.
FR3	Detect and localize suspected leaks.	Produce reviewable incidents and location estimates.
FR4	Isolate confirmed breaks under approved rules.	Obey permissions, constraints, and expiry conditions.
FR5	Verify physical valve response.	Never record unconfirmed closure as successful isolation.
FR6	Coordinate bounded pressure adjustments.	Remain within approved limits and adjustment rates.

4.7 Non-Functional Requirements:

Performance: Evaluate 5,000 simulated nodes transmitting once per second as a proposed benchmark, not an established production capacity. Assess acoustic bandwidth separately. Availability: Provide redundant critical services and approved recovery targets. Security: Authenticate users and devices, encrypt communications, protect keys, and validate commands. Usability: Display freshness, severity, uncertainty, assets, and control status. Maintainability: Document interfaces and version configurations. Integrity: Preserve original observations and trace corrections.

4.8 Software Engineering Principles:

Modularity separates ingestion, analysis, control, and reporting to limit change impact. Scalability partitions workloads by zone and expands capacity after testing. Reliability uses durable messaging, health monitoring, and degraded modes. Maintainability uses configurable rules and versioned changes. Usability explains incident causes and actions. Security applies least privilege and separates control from administration. Traceability links decisions to evidence and configuration.

4.9 Fault-Tolerant Behaviour:

Gateways retain unacknowledged events within storage limits. Services replay and deduplicate using device identity and sequence numbers. Physical commands require unique IDs, expiry times, authorization, and state verification. A timed-out command is not blindly repeated because its effect may already have occurred. During partitions, valid control authority must prevent competing commands. Loss of safe control conditions triggers a defined degraded mode. Audit protection combines append-only access, hash-linked records, signed checkpoints, and separately protected copies. These controls detect tampering but cannot prove the physical accuracy of every sensor observation.

5. Justify the Chosen Methodology or Framework:

5.1 Selected Approach:

Use Agile Scrum with DevOps practices and engineering approval gates. Scrum supports iterative delivery and stakeholder feedback. DevOps supports repeatable builds, automated checks, controlled deployment, and monitoring. Engineering gates govern physical-control capabilities.

5.2 Justification Against Alternatives:

Approach	Suitability assessment
Waterfall alone	Structured documentation, but late feedback may expose integration and detection problems after substantial development.
Agile without control gates	Supports adaptation but does not independently establish safe field-release conditions.
Selected approach	Combines early feedback, repeatable verification, documented evidence, and controlled operational release.

The selected approach accommodates uncertain interfaces and detection behaviour while retaining stronger approval for actions affecting public supply.

5.3 Development Plan:

Increment	Deliverable and completion evidence
Requirements baseline	Stakeholders, constraints, assumptions, risks, and acceptance criteria; reviewed by utility engineers.
Telemetry foundation	Registry, simulator, ingestion, dashboard; verified recovery and data quality.
Detection and localization	Correlation, acoustic analysis, confidence; evaluated against labelled scenarios.
Control and audit	Valve simulation, authorization, feedback, audit chain; control-failure and integrity checks.
Integrated pilot	Resilience, pressure limits, reports, procedures; observation-mode and supervised acceptance.

The Product Owner prioritizes needs; developers implement; operators and engineers review. The Definition of Done includes documentation, verification, traceability, and review.

5.4 Grouped Module Testing:

Test group	Scenario and pass condition
Ingestion and recovery	Reconnect a gateway within retention limits; recover events without duplicate utilization counts.
Detection and localization	Replay leaks, demand changes, and noise; meet agreed criteria and flag uncertainty.
Control and pressure	Test stale commands, stuck valves, and protected areas; block unsafe actions and escalate.
Resilience and security	Fail a service and attempt unauthorized control; recover correctly and reject access.
Audit and reporting	Alter a copied record and reconcile totals; detect changes and retain traceability.

Test status: Not executed. These are planned acceptance tests, not implementation results.

5.5 Maintenance:

Review false alarms, field-confirmed locations, failures, and recovery events. Update the backlog and validate changes in simulation and observation mode before release. Maintain software and configuration rollback procedures.

6. Discuss Expected Outcomes and Limitations

6.1 Expected Benefits and Evaluation:

Expected benefit	Evaluation method
Reduced water loss	Compare incident discharge estimates and isolation duration with the baseline.
Faster investigation	Measure alert-to-field-confirmation time.

Dependable response	Measure verified isolation time and unsuccessful command frequency.
Better pressure management	Review pressure-band compliance and control stability.
Accountability	Check incident reconstruction and utilization reconciliation.

Numerical improvement claims require pilot evidence and comparable operating conditions. These benefits are expected, not demonstrated results.

6.2 Risks and Mitigation:

Sensor drift requires calibration and health checks. Inaccurate topology requires field verification before automation. Outages require buffering, capacity monitoring, and gap reporting. False alarms require correlated evidence and tested thresholds. Stuck valves require feedback, escalation, and reviewed alternatives. Security risks require restricted access, protected credentials, monitoring, and response procedures.

Deployment also involves equipment costs, training, integration effort, and ongoing maintenance. A phased pilot limits exposure while identifying practical constraints.

6.3 Practical Case Scenario:

In a hypothetical incident, pressure drops while flow rises. Two synchronized acoustic sensors identify a consistent disturbance, unrelated to scheduled operations. For an illustrative straight segment, L = 400 metres, assumed c = 1,000 metres per second, and $\Delta t = -0.10$ seconds:

$$\mathbf{x = [400 + (1{,}000 \times -0.10)]/2 = 150 \text{ metres}}$$

The estimated position is 150 metres from sensor A, subject to uncertainty. The platform checks isolation and protected-service rules, sends approved commands, and verifies feedback. A failed valve triggers escalation rather than a success claim. Field teams inspect the area and record the actual fault location.

6.4 Future Enhancements:

Possible enhancements include calibrated hydraulic simulation, improved acoustic interpretation, battery-aware sampling, mobile inspection, and maintenance-system integration. Predictive analysis could prioritize vulnerable sections if sufficient reliable historical data becomes available.

6.5 Conclusion:

The platform covers dependable collection, evidence-based detection, localization, controlled isolation, verification, and documented restoration. Fault tolerance supports continuity during defined failures; modular design and traceable decisions support maintenance and auditability. Scrum, DevOps, and engineering gates provide a practical development approach.

Deployment depends on verified asset information, calibrated devices, validated rules, trained personnel, and measured pilot results. Together, these measures provide a credible foundation for reducing water loss and improving municipal response.